

Question Number	Answer	Additional Guidance	Mark
4a	A description that includes two of the following points: • find a suitable {mass / length} of plant to use (1) • find a suitable method to change light intensity (1) • find a suitable {temperature / time} to collect {gas / O₂} (1) • find suitable {method / apparatus} to measure volume of {gas / O₂} (1)	A method to provide quantitative results Collection of gas / O₂ needs only be mentioned only once Allow suitable range of light intensities Ignore wavelength Allow optimum temperature Ignore number of bubbles rate of photosyn	2

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4b	An answer that includes nine of the following points: • clear statement of the dependent variable e.g. the volume of {gas / O₂} released (1) • description of method of measuring volume of gas (1) • method of standardising plants (1) • method of producing two different light intensities (1) • use of (sodium) hydrogencarbonate (1) • allow plant time to acclimatise (1) • {standardised / stated} time for gas collection (1) • one variable that need to be controlled and its method of control (1) • {repeats / repeat the whole experiment} to give mean and SD (1) • method of calculation of rate (1)	ALLOW different valid methods. Do not allow carbon dioxide Ignore reference to time Allow (labelled) diagram / photosynthometer Allow length / mass / preconditioning Ignore age / species / size eg moving lamp / wattage of bulb / darkened room and light room Allow 5 mins to 24 hrs eg temperature – TC waterbath / heatshield / LED bulb Ignore AC room wavelength of light – filter / bulb pH – buffer eg $\frac{\text{volume}}{\text{time}}$ / $\frac{\pi r^2 d}{t}$ / gradient of graph	9

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4c	An answer that includes the following points: • table for raw data with headings and units, with means calculated from repeats (1) AND • bar graph format with labelled axes (1) • use of a t test for difference (with a bar graph showing only two different light intensities) OR • {line / scatter} graph format with labelled axes (1) • use of a correlation test difference (with a line graph showing more than two different light intensities)	Allow a description of how to calculate mean in text or as part of an axis label Allow other valid tests Allow named test e.g. Spearmans Rank	3

Question Number	Answer	Additional Guidance	Mark
4d	An answer that includes two of the following points: • difficult to measure (small) {volumes / distances} of {gas / O₂} (with precision) (1) • difficult to control temperature (1) • difficult to control surface area of leaves (1) • idea that actual growing conditions in the river change during the day so the results may not be a fair representation (1)	Allow {some O₂ dissolved / some O₂ used in respiration / some gas may be CO₂} Ignore bubbles / escape of gas / errors in measuring Allow example of change	2

Total for question 4 = 16

